

Layer 1: SNC Detection

Active Set
Thresholding

26-Neighbor
BFS Labeling

Field-Weighted
Centroid

SNC positions

Layer 2: Route Computation

SNC Mesh
Construction

Dijkstra
All-Pairs SP

Mixed Objective
Assignment

routing tables

Layer 3: Message Distribution

LOAD_REPORT
(Sat $\rightarrow$ SNC)

CORE_TOPO
(SNC $\leftrightarrow$ SNC)

ROUTE_DIST
(SNC $\rightarrow$ Sat)

CORE_ASSIGN

message triggers

Layer 4: State Machine (per satellite)

INIT

MEMBER

CENTER

RELAY

RECONFIG